

Integrality of the sequence OEIS A276175

Abstract

We prove that the order-four rational recurrence

$$a_{n+4} a_n = (a_{n+1} + 1)(a_{n+2} + 1)(a_{n+3} + 1), \quad a_0 = a_1 = a_2 = a_3 = 1,$$

produces only integers. A rational number is an integer iff it has nonnegative valuation at every prime, so the statement splits into two independent parts. For odd primes the valuations are controlled by an elementary “atom” induction. The remaining obstruction, at the prime 2, is settled by a 44-step 2-adic return lemma for rational points of the tube $1 + 16\mathbb{Z}_2$, whose computational core is a finite, decidable check in the rings $\mathbb{Z}/2^m\mathbb{Z}$.

1 Statement and strategy

Let $(a_n)_{n \geq 0}$ be defined by

$$a_0 = a_1 = a_2 = a_3 = 1, \quad a_{n+4} a_n = (a_{n+1} + 1)(a_{n+2} + 1)(a_{n+3} + 1) \quad (n \geq 0). \quad (1)$$

A straightforward induction shows every a_n is a well-defined positive rational (each $a_n > 0$, so no division by zero occurs).

Theorem 1. $a_n \in \mathbb{Z}$ for every $n \geq 0$.

A nonzero rational r is an integer if and only if $v_p(r) \geq 0$ for every prime p , where v_p is the p -adic valuation. Since each a_n is a positive rational, Theorem 1 is equivalent to

$$v_p(a_n) \geq 0 \quad \text{for every prime } p \text{ and every } n \geq 0. \quad (2)$$

We prove (2) in two parts, treated by completely different methods:

- Odd primes p (Section 2): an elementary induction on auxiliary integers $t_p(n)$ attached to the shifted sequence $b_n = a_n + 1$. This yields $a_n \in \mathbb{Z}[\frac{1}{2}]$.
- The prime 2 (Section 3): a return lemma for rational 2-adic tube points, yielding an explicit nonnegative 44-periodic word.

Together they give (2) and hence Theorem 1 (Section 4).

Throughout, write $b_n = a_n + 1$. From (1), $a_{n+4}a_n = b_{n+1}b_{n+2}b_{n+3}$, and expanding $b_{n+4} = a_{n+4} + 1$ gives the identity used repeatedly below:

$$b_{n+4} a_n = a_n + b_{n+1}b_{n+2}b_{n+3}. \quad (3)$$

All a_n, b_n are positive, hence nonzero, so every valuation of a product or quotient below splits additively.

2 Odd primes: an atom induction

Fix an odd prime p and abbreviate $v_p = v$. Define integers $t_n := t_p(n)$ by

$$t_0 = t_1 = t_2 = t_3 = 0, \quad t_{n+4} = v(b_{n+4}) - t_n, \quad (4)$$

so that, by construction,

$$v(b_{n+4}) = t_{n+4} + t_n. \quad (5)$$

Thus t_n is the “atom” of p -adic valuation freshly deposited at index n ; (5) says the valuation of b_{n+4} is the sum of the atom it carries and the atom four steps back.

The valuation identity is established first; it does not use nonnegativity of the atoms.

Lemma 1 (Valuation identity). *For every $n \geq 0$,*

$$v(a_{n+4}) = t_{n+1} + t_{n+2} + t_{n+3}. \quad (6)$$

Proof. We use strong induction in the same form as the Lean lemma `val_a`. For $n = 0, 1, 2, 3$, use

$$a_4 = 2^3, \quad a_5 = 2^2 \cdot 3^2, \quad a_6 = 2 \cdot 3^2 \cdot 37, \quad a_7 = 3^2 \cdot 37 \cdot 667,$$

and

$$t_4 = v(9), \quad t_5 = v(37), \quad t_6 = v(667).$$

Because p is odd, $v(2) = 0$, so these four cases give (6) directly.

For the induction step write $n = m + 4$. Taking valuations in the recurrence at index $m + 4$ and using (5) gives

$$\begin{aligned} v(a_{m+8}) + v(a_{m+4}) &= (t_{m+5} + t_{m+1}) + (t_{m+6} + t_{m+2}) \\ &\quad + (t_{m+7} + t_{m+3}). \end{aligned}$$

The induction hypothesis at m is $v(a_{m+4}) = t_{m+1} + t_{m+2} + t_{m+3}$. Subtraction proves $v(a_{m+8}) = t_{m+5} + t_{m+6} + t_{m+7}$, as required. $\square$

Proposition 1 (Nonnegativity and burst confinement). *For every $n \geq 0$ the following hold:*

(I) $t_n \geq 0$;

(III) *for every k such that $n = k + 4$, if $t_n > 0$, then $t_{k+1} = t_{k+2} = t_{k+3} = 0$.*

Property (III) says that positive atoms occur only in isolated bursts. Notice that Lemma 1 is already available before (I) and (III) are proved.

Proof. We prove (I) and (III) together by strong induction, exactly as in the Lean lemma `main_induction`. The base indices $n = 0, \dots, 7$ use $t_0 = \dots = t_3 = 0$ and

$$b_4 = 3^2, \quad b_5 = 37, \quad b_6 = 23 \cdot 29, \quad b_7 = 2^5 \cdot 11 \cdot 631,$$

so $t_j = v(b_j) \geq 0$ for $4 \leq j \leq 7$. If one is positive, primality of p and the displayed factorizations force the preceding three atoms to vanish; for example, $t_6 > 0$ forces $p = 23$ or 29 , neither of which divides b_4 or b_5 . This checks both assertions at the eight base indices.

For the induction step, assume both assertions through index $k + 7$ and prove them at $k + 8$. First prove (I). Lemma 1 and the induction hypotheses give

$$v(a_{k+8}) = t_{k+5} + t_{k+6} + t_{k+7} \geq 0.$$

We need $v(b_{k+8}) \geq t_{k+4}$, which is equivalent to $t_{k+8} \geq 0$ by (4). If $t_{k+4} = 0$, then $b_{k+8} = 1 + a_{k+8}$ and the ultrametric inequality immediately gives $v(b_{k+8}) \geq 0$.

Suppose instead that $T := t_{k+4} > 0$. The already proved burst assertion at index $k+4$ gives $t_{k+1} = t_{k+2} = t_{k+3} = 0$. Hence Lemma 1 gives

$$v(a_{k+4}) = t_{k+1} + t_{k+2} + t_{k+3} = 0.$$

Rewrite (3) at index $k+4$ as

$$b_{k+8}a_{k+4} = b_{k+4} + a_{k+7}b_{k+6}b_{k+5} + a_{k+6}b_{k+5} + a_{k+5}. \quad (7)$$

Every summand on the right has valuation at least T . Indeed,

- $v(b_{k+4}) = t_{k+4} + t_k \geq T$;
- $v(a_{k+5}) = t_{k+2} + t_{k+3} + t_{k+4} = T$;
- $v(a_{k+6}) = t_{k+3} + t_{k+4} + t_{k+5} \geq T$ and $v(b_{k+5}) = t_{k+5} + t_{k+1} \geq 0$;
- $v(a_{k+7}) = t_{k+4} + t_{k+5} + t_{k+6} \geq T$, while $v(b_{k+5}), v(b_{k+6}) \geq 0$.

Thus (7) has valuation at least T . Since $v(a_{k+4}) = 0$, we obtain $v(b_{k+8}) \geq T$, proving $t_{k+8} \geq 0$.

It remains to prove (III) at $k+8$. If $t_{k+8} > 0$, then

$$v(b_{k+8}) = t_{k+8} + t_{k+4} > 0$$

by the nonnegativity already known through $k+7$. Since $a_{k+8} = b_{k+8} - 1$, it follows that $v(a_{k+8}) = 0$. Applying Lemma 1 at $k+4$ yields

$$0 = v(a_{k+8}) = t_{k+5} + t_{k+6} + t_{k+7}.$$

All three atoms are nonnegative by the induction hypothesis, so each is zero. This proves the burst assertion at $k+8$ and closes the induction. $\square$

Corollary 2. $v(a_n) \geq 0$ for all n . As p ranges over all odd primes, $a_n \in \mathbb{Z}[\frac{1}{2}]$.

Proof. For $n \leq 3$, $a_n = 1$ and $v(a_n) = 0$. For $n \geq 4$, Lemma 1 and Proposition 1 give $v(a_n) = t_{n-3} + t_{n-2} + t_{n-1} \geq 0$. Hence no odd prime divides the denominator of a_n . $\square$

3 The prime 2: a 44-step return

The behaviour at $p = 2$ is genuinely 2-adic and does not reduce to a congruence in a fixed modulus: naive term-by-term 2-adic estimates lose precision at every step. The remedy is to control an entire 2-adic neighbourhood of the orbit simultaneously and to exhibit an explicit period.

Let

$$w = (w_0, \dots, w_{43}), \quad t = (t_0, \dots, t_{43})$$

be the 44-periodic words

$$\begin{aligned} w &= (0, 0, 0, 0, 3, 2, 1, 0, 2, 3, 4, 0, 0, 0, 0, 3, 2, 1, 0, 2, 3, 4, \\ &\quad 0, 0, 0, 0, 4, 3, 2, 0, 1, 2, 3, 0, 0, 0, 0, 4, 3, 2, 0, 1, 2, 3), \\ t &= (1, 1, 1, 1, 0, 0, 0, 5, 0, 0, 0, 2, 1, 1, 1, 0, 0, 0, 5, 0, 0, 0, \\ &\quad 2, 1, 1, 2, 0, 0, 0, 5, 0, 0, 0, 1, 1, 1, 2, 0, 0, 0, 5, 0, 0, 0), \end{aligned}$$

extended 44-periodically to all indices. A direct check gives the *balance law*

$$w_{n+4} + w_n = t_{n+1} + t_{n+2} + t_{n+3} \quad (n \geq 0). \quad (8)$$

Lemma 2 (44-step rational return lemma). *Let*

$$A_0, A_1, A_2, A_3 \in \mathbb{Q} \cap (1 + 16\mathbb{Z}_2)$$

and extend the rational sequence (A_n) by

$$A_{n+4}A_n = (1 + A_{n+1})(1 + A_{n+2})(1 + A_{n+3}).$$

Then $A_0, \dots, A_{47}$ are all defined and nonzero, with

$$v_2(A_n) = w_n \quad (0 \leq n \leq 43),$$

and moreover

$$A_{44}, A_{45}, A_{46}, A_{47} \in \mathbb{Q} \cap (1 + 16\mathbb{Z}_2).$$

The restriction to rational points is deliberate: this is the domain quantified over in the Lean development. Equivalently, write $A_i = 1 + 16c_i$ with $c_i \in \mathbb{Q}$ and either $c_i = 0$ or $v_2(c_i) \geq 0$. The recurrence stays in $\mathbb{Q}$. The last conclusion therefore has exactly the same form as the hypothesis, so the lemma can be iterated in blocks of 44.

Proof. The normalized recurrence motivates the finite computation. Put $S_n = A_n/2^{w_n}$ and

$$H_n := \frac{1 + A_n}{2^{t_n}}.$$

Whenever the displayed divisions have the predicted valuations, the balance law (8) turns the original recurrence into

$$S_{n+4} = S_n^{-1} H_{n+1} H_{n+2} H_{n+3}. \quad (9)$$

The formal proof does not construct infinite series in $\mathbb{Z}_2[[z_0, z_1, z_2, z_3]]$. Instead, for each $0 \leq n \leq 47$ the finite checker constructs an integer polynomial $P_n(z_0, z_1, z_2, z_3)$ and a precision p_n . The soundness invariant, called **Rep** in Lean, is

$$A_n \neq 0, \quad v_2(A_n) = w_n, \quad v_2\left(\frac{A_n}{2^{w_n}} - P_n(c)\right) \geq p_n. \quad (10)$$

Here exact equality in the last comparison is allowed, as usual for a 2-adic approximation. Thus the polynomial jet represents the actual normalized rational value uniformly for every rational tube parameter c .

The initial jets are $P_i = 1 + 16z_i$ at precision $p_i = 34$. At each step the checker performs finite polynomial addition and multiplication, exact division by the required power of 2, and Newton inversion modulo 2^{p_n} . It also checks the valuation-reading conditions. When $w_n = 0$, these are

$$v_2(1 + P_n(0)) = t_n, \quad \mu_n := \min_{\alpha \neq 0} v_2([z^\alpha]P_n) \geq t_n + 1. \quad (11)$$

They imply $v_2(1 + A_n) = t_n$ for every admissible rational c . When $w_n > 0$, the checker verifies $t_n = 0$, and $1 + A_n$ is a 2-adic unit. Finally, at $n = 44, 45, 46, 47$ it checks

$$P_n(0) \equiv 1 \pmod{16}, \quad \mu_n \geq 4,$$

together with $p_n \geq 4$.

The finite Boolean computation alone is not the proof of the rational statement. Its soundness bridge is the Lean lemma **step_prep**: assuming (10) at $n, n+1, n+2, n+3$, it proves (10) at $n+4$ using (9) and the checked polynomial operations. Starting with the four initial jets, the lemma **reps** applies this step through index 47. Consequently the first two assertions of (10) give nonvanishing and the valuation word over the first block. At the four return indices,

the last assertion of (10), the congruence and depth checks above, and $p_n \geq 4$ show $A_n \equiv 1 \pmod{16\mathbb{Z}_2}$. This is formalized by `return_tube`; `seq_offset` identifies the sequence generated by the returned window with the original sequence shifted by 44.

For orientation, the decisive phases with $w_n = 0$ give the following checked constant terms and nonconstant depths:

n	0	1	2	3	7	11	12	13	14	18	22	23
t_n	1	1	1	1	5	2	1	1	1	5	2	1
$P_n(0) \bmod 64$	1	1	1	1	31	43	9	21	9	31	11	45
μ_n	4	4	4	4	6	3	4	4	3	7	4	4

n	24	25	29	33	34	35	36	40	44	45	46	47
t_n	1	2	5	1	1	1	2	5	1	1	1	1
$P_n(0) \bmod 64$	61	43	31	17	21	41	3	31	17	49	17	49
μ_n	4	4	7	3	4	4	3	6	4	4	4	4

The running precisions at $P_{44}, P_{45}, P_{46}, P_{47}$ are respectively 9, 8, 7, 6, above the four bits required for the return congruence. The checker starts from precision 34 and is discharged in Lean by `native_decide`. $\square$

Corollary 3. *For every $n \geq 0$,*

$$v_2(a_n) = w_n = w_{n \bmod 44} \geq 0.$$

Proof. The initial rational window $a_0 = a_1 = a_2 = a_3 = 1$ lies in $\mathbb{Q} \cap (1 + 16\mathbb{Z}_2)$, and (a_n) obeys the recurrence of Lemma 2. That lemma gives the valuation word for the first block and returns $(a_{44}, a_{45}, a_{46}, a_{47})$ to the same rational tube. The shifted recurrence is the same recurrence, so iteration from indices 44, 88, $\dots$ proves the formula. Every entry of w is nonnegative. $\square$

The supplied Lean file packages the block induction needed by the integrality theorem as `Return.val_nonneg`, and exposes $v_2(a_n) \geq 0$ as `val_nonneg_a`. It does not package the exact all- n equality of Corollary 3 as a separate named theorem. That equality is the paper-level consequence of the same proved first-block invariant `reps`, return lemma `return_tube`, and shift lemma `seq_offset`.

4 Conclusion

The formal dependency chain for integrality is

$$\begin{aligned}
& \text{odd-prime valuation induction} \\
& + \text{44-step rational 2-adic return} \\
& + \text{the valuation criterion for rational integers} \\
& \implies \text{integrality.}
\end{aligned}$$

Neither the cluster embedding nor the Laurent phenomenon is used in this chain.

Proof of Theorem 1. By Corollary 2, $a_n \in \mathbb{Z}[\frac{1}{2}]$: no odd prime divides its denominator. By the machine-checked nonnegativity consequence described after Corollary 3, $v_2(a_n) \geq 0$, so no power of 2 divides its denominator either. Hence a_n has denominator 1, and therefore $a_n \in \mathbb{Z}$. $\square$

Remark (Optional context: the cluster embedding). The Lean file proves the following embedding, but the final integrality theorem does not use it. Define (x_n) by

$$x_0 = 4, \ x_1 = \frac{1}{2}, \ x_2 = 4, \ x_3 = \frac{1}{2}, \quad x_{n+4} x_n = 1 + x_{n+1} x_{n+2} x_{n+3}.$$

Then (a'_n) with $a'_n := x_{n+1}x_{n+2}x_{n+3}$ satisfies (1) with the same four initial values, and the formal lemma **cluster** proves $a_n = x_{n+1}x_{n+2}x_{n+3}$.

The further identification of the x -recurrence as a period-one cluster recurrence and the invocation of the Laurent phenomenon are external context, not results proved in the supplied Lean file. They give another mathematical route to $x_n \in \mathbb{Z}[\frac{1}{2}]$ and hence $a_n \in \mathbb{Z}[\frac{1}{2}]$, but they are not part of the proof dependency chain above.

Remark. Corollary 3 shows that the valuation sequence is 44-periodic. A direct finite comparison of the displayed word shows that its least period is 44; this minimality observation is not separately formalized in the supplied Lean file and is not needed for integrality.